SAI PRASANNA GUJJULA

+1 945 296 0036 | saiprassu16@gmail.com|Linkedin

SUMMARY

Detail-oriented Full Stack Developer and AI/ML enthusiast with 3+ years of experience delivering scalable web applications and intelligent systems using Python, React.js, Flask/Django, .NET, and modern DevOps practices. Proven ability to design and deploy microservices, implement secure RESTful APIs, and optimize CI/CD pipelines using Docker, GitHub Actions, and Jenkins. Skilled in building responsive, data-driven UIs and integrating machine learning models for real-time inference in domains like finance and retail. Demonstrated success in leading full project cycles—from database design and frontend development to model training and deployment. Hands-on experience with deep learning, computer vision, and evaluation metrics. Also possess a foundational understanding of Generative AI, including prompt-based workflows and vector embeddings, with a keen interest in further exploration.

TECHNICAL SKILLS

- **Languages & Scripting**: Python, JavaScript, Go,TypeScript, HTML5, CSS3, Shell Scripting, SQL, C#
- **Frontend &Backend Frameworks:** React.js, Redux, Bootstrap, Flask, Django, Node.js, .NET (C#), REST APIs, GraphQL
- **Databases:** PostgreSQL, MySQL, MongoDB, Redis
- **DevOps & CI/CD**: GitHub Actions, Jenkins, GitFlow, Infrastructure as Code (IaC), SonarQube, YAML, OAuth2
- **Containerization & Deployment**: Docker, Docker Compose, Kubernetes (basic), AWS (Lambda, Elastic Beanstalk)
- **Testing & QA:** PyTest, Postman, Unit Test, Integration Testing, Cypress (basic), Manual QA coordination
- **AI/ML & Data Processing**: TensorFlow, Scikit-learn, OpenCV, Pandas, NumPy, GenAI (Embeddings, Transformers), Matplotlib
- **Tools & Collaboration**: Git, Jira, Confluence, VS Code, Postman, Slack, Agile/Scrum
- **Other Skills:** Problem Solving, Decision Making, Analytical Thinking, Debugging, Requirements Analysis, Cross-functional Collaboration, System Design, Documentation

EXPERIENCE

CITI Bank | Full stack developer **Jun2024 – May2025**

- Developed and deployed 20+ reusable React.js components and scalable Flask-based REST APIs to power banking dashboards and internal financial workflows, ensuring modularity and responsiveness.
- Built and optimized PostgreSQL-integrated microservices, resolving production issues and improving query efficiency across real-time transactional systems, while adhering to clean code principles and design patterns.
- Architected secure, audit-compliant CI/CD pipelines using Docker, GitHub Actions, and Jenkins, automating AWS Lambda deployments, managing secrets via (Infrastructure as Code)IaC, and reducing manual effort by 40%.
- Standardized DevOps workflows with reusable pipeline templates, container strategies, and integrated monitoring/alerting, enabling faster onboarding, proactive troubleshooting, and alignment with compliance standards.
- Conducted peer code reviews and implemented automated testing using PyTest and Postman, maintaining 85%+ test coverage and ensuring stability across Flask-based APIs and data pipelines.
- Integrated Kafka-based event-driven architecture into Flask microservices to enable real-time ingestion and processing of financial transactions, decoupling services and reducing API latency by 30%.

DXC Technologies | Associate software Engineer **May2021 - Jul2023**

- Developed and maintained full-stack modules using React.js and Django to support inventory, order tracking, and customer operations across 161 retail store locations.
- Built RESTful APIs with Python, PostgreSQL, and MongoDB, enabling real-time data sync between distributed systems and centralized services while maintaining reliability and scalability.
- Automated CI/CD workflows using GitHub Actions, Jenkins, and Docker, and adopted PaaS strategies (AWS Beanstalk/Azure App Services) to simplify deployment, scaling, and rollback processes.
- Enforced clean code practices and GitFlow branching, collaborated in Agile sprints via Jira, and improved test coverage with PyTest to reduce bugs and accelerate release cycles.
- Instrumented backend observability by integrating Python logging and configuring alerts with AWS CloudWatch and Jenkins notifications, enabling proactive error detection and faster issue resolution.

Verzeo EdTech | AI/ML Intern **Mar 2020-Aug 2020**

- Designed and deployed Flask-based machine learning inference APIs that served real-time predictions for 500+ inputs, integrating HTML/CSS dashboards to simulate complete ML workflows for demos and testing.
- Explored and prototyped GenAI techniques using transformers and vector embeddings, integrating prompt-based inference into pipelines and demonstrating use cases across 3 educational projects.
- Preprocessed and modeled over 10,000 facial images and structured records using NumPy, Pandas, Scikit-learn, and OpenCV to train models with 15% improvement in validation accuracy.
- Containerized Python applications with Docker, implemented test-driven evaluation (accuracy, precision, confusion matrix), and improved inference stability across multiple deployment iterations with 20% reduction in error rate.

ACADEMIC PROJECTS

Log Tracking Application:

- Designed a full-stack health tracker using React and .NET, enabling users to log food intake, fitness metrics, and daily health parameters with persistent tracking features.
- Built 15+ reusable frontend components and 10+ backend RESTful APIs, supported by a normalized 12-table PostgreSQL schema for consistent and scalable data storage.
- Implemented secure token-based authentication, allowing protected user sessions and personalized dashboards with real-time health insights across devices.
- Enhanced user experience with dynamic health visualizations and mobile-responsive design, integrating interactive charts, progress tracking, and conditional rendering using styled components and React hooks.

Deep Learning methods to detect SARS-CoV-2 using chest X-ray

- Designed and trained CNN-based classifiers using chest X-ray images, leveraging VGG16, VGG19, ResNet, DenseNet, and Inception architectures to detect SARS-CoV-2 with high accuracy.
- Fine-tuned pre-trained Keras models with ImageNet weights, applied data augmentation to balance small datasets, and achieved 99.8% accuracy with VGG16 on COVID-19 prediction.
- Developed an interactive Flask-based frontend where users could select the model, upload an X-ray image, and receive real-time classification results for COVID-19 or normal cases.
- Built an end-to-end ML pipeline with Conv, ReLU, pooling, flattening, and sigmoid layers, and evaluated performance using accuracy, recall, precision, and F1-score metrics.

Age detection with facial features

- Developed Flask-based microservices to serve age prediction models (CNN, SVM, KNN) through secure REST APIs, enabling real-time facial age classification.
- Labeled and preprocessed over 1,000 facial images using OpenCV and NumPy, enhancing model consistency and improving prediction accuracy through structured input transformation.
- Connected PyTest-validated backend with React-based UI, reducing classification error by 10% and ensuring reliable results through modular validation and error-handling workflows.
- Designed a full pipeline from image upload to prediction output, enabling efficient and scalable age inference through asynchronous request handling and user feedback mechanisms.
- Gained foundational knowledge of Generative AI concepts, including prompt engineering and embeddings, and explored potential integration with transformer models for vision tasks.

EDUCATION

University of North Texas, Denton, TX **Aug2023 - May2025**
Master of Science ,Computer and Information Sciences

Sridevi Women's Engineering College ,Hyderabad **Jul2018 – Aug2022**
Bachelors of Technology ,Electronics and Communication